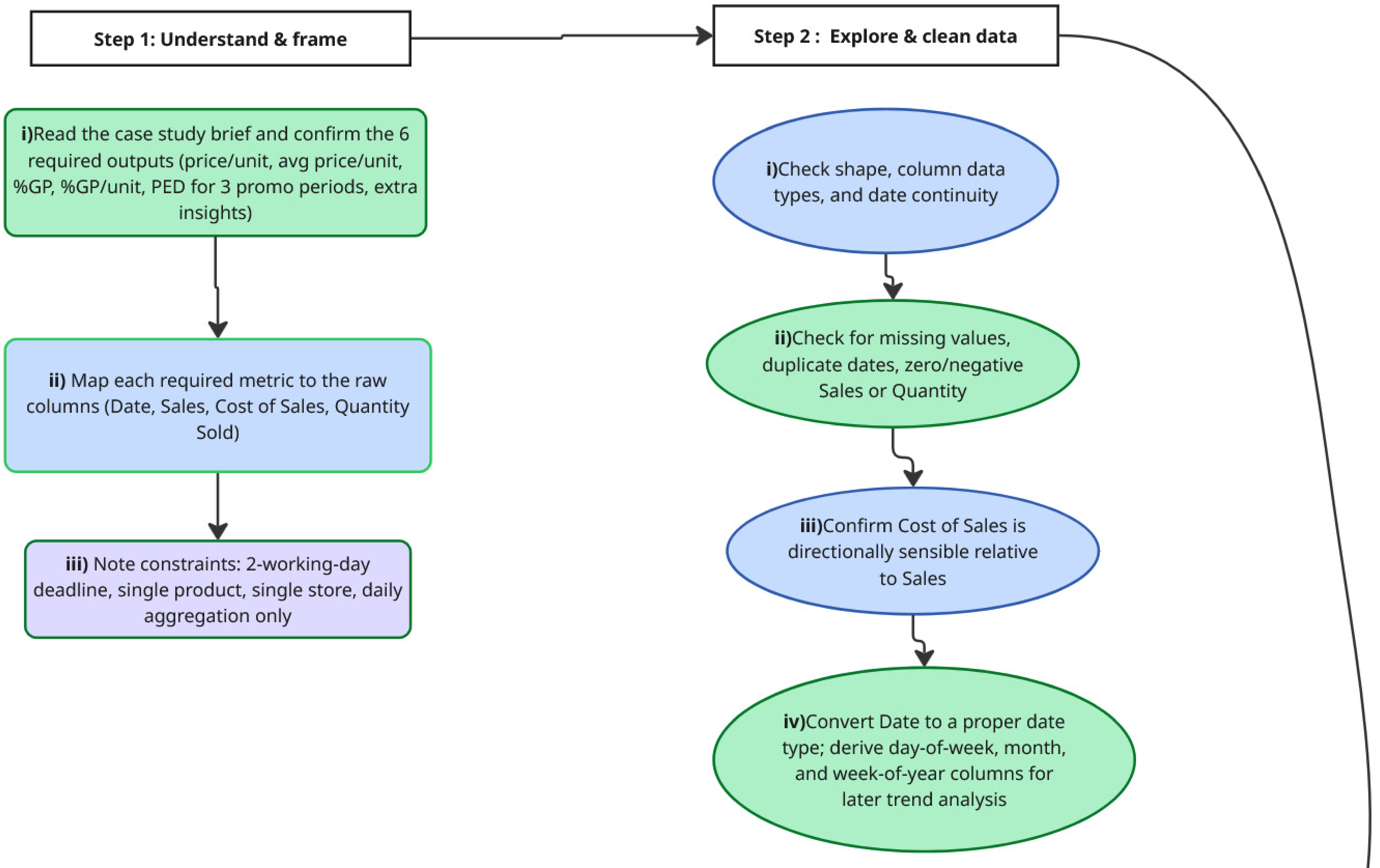
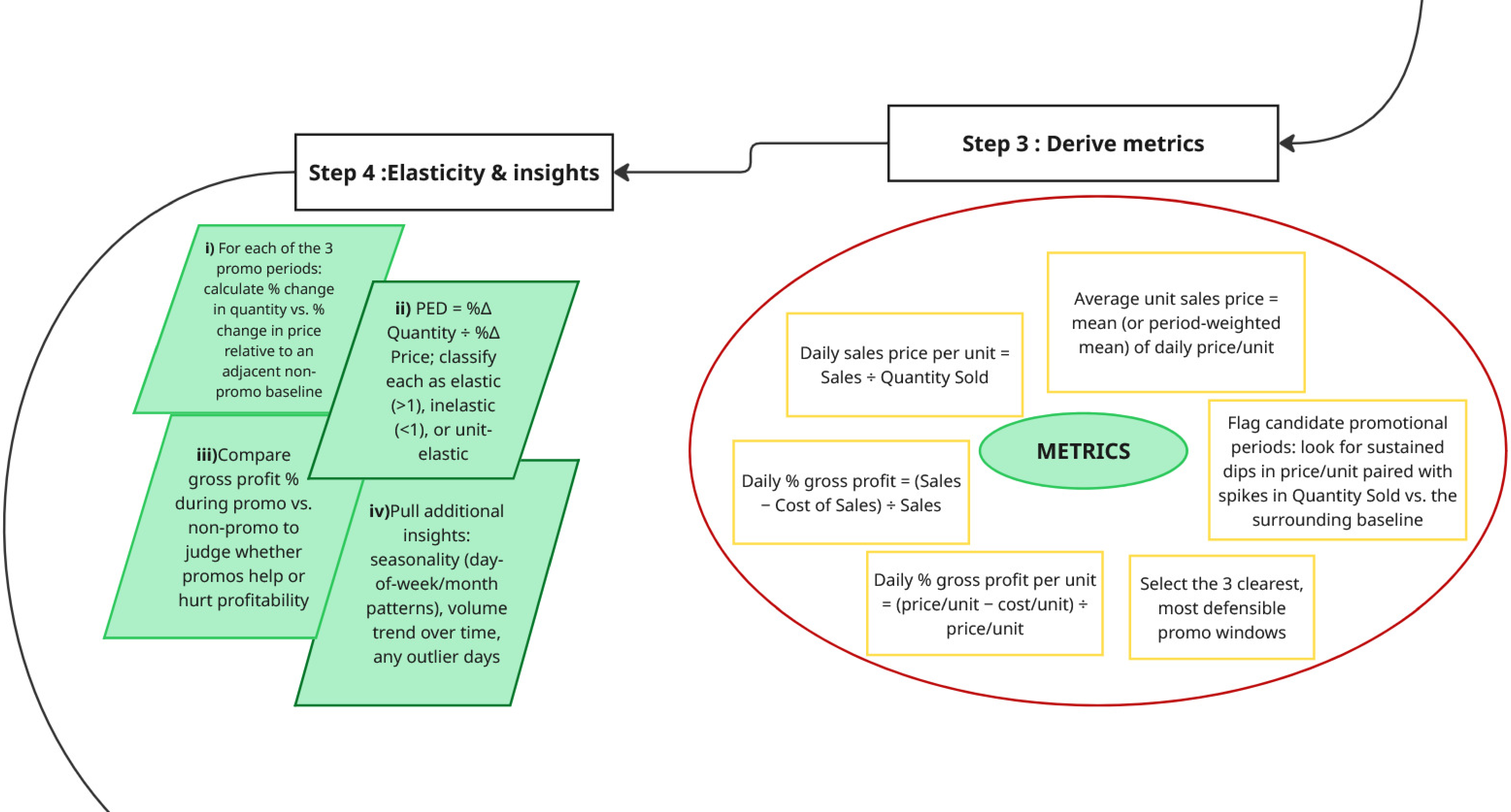
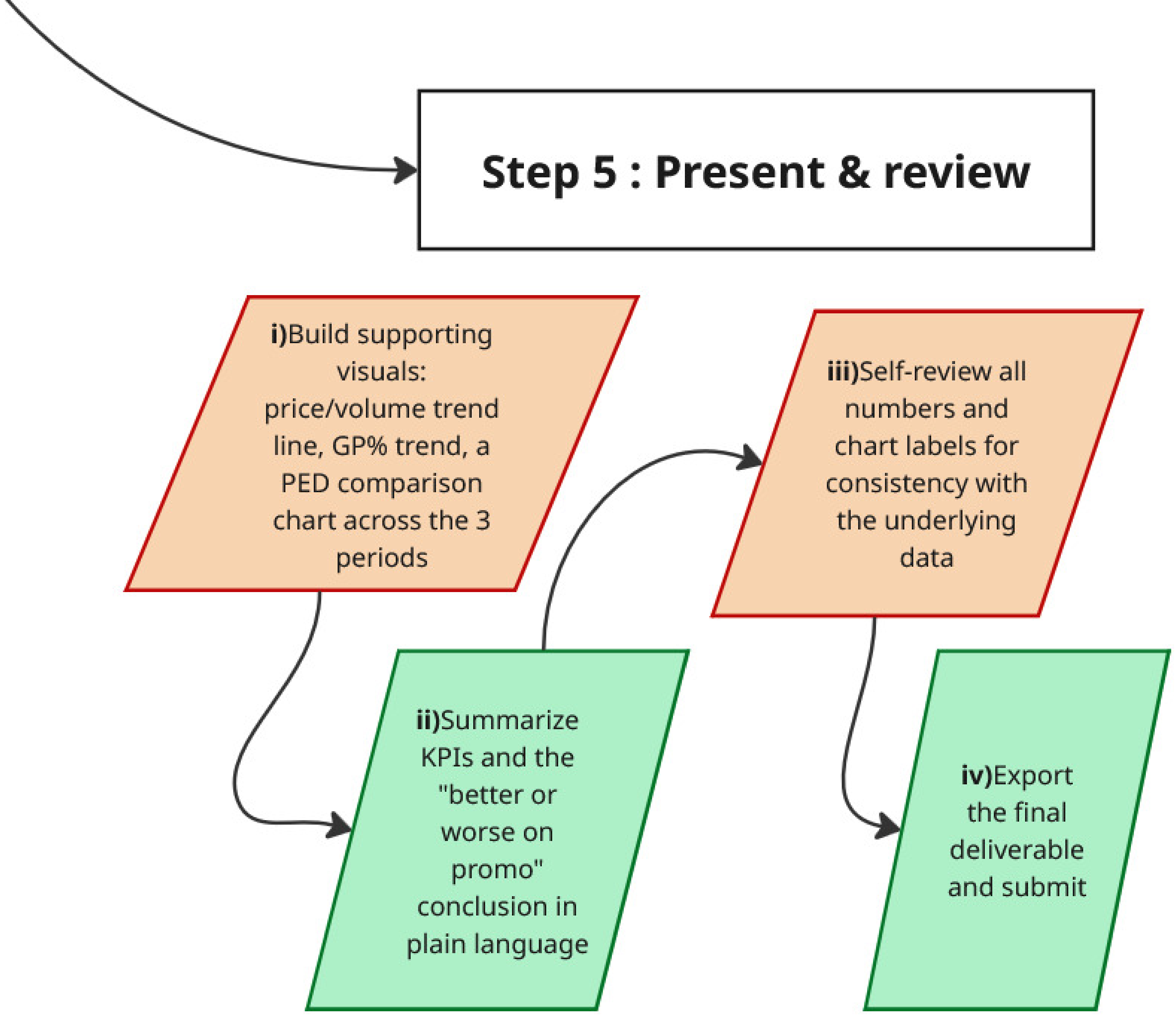

Step 1: Understand & frame
i)Read the case study brief and confirm the 6 required outputs (price/unit, avg price/unit, %GP, %GP/unit, PED for 3 promo periods, extra insights)
ii) Map each required metric to the raw columns (Date, Sales, Cost of Sales, Quantity Sold)
iii) Note constraints: 2-working-day deadline, single product, single store, daily aggregation only
Step 2 : Explore & clean data
i)Check shape, column data types, and date continuity
ii)Check for missing values, duplicate dates, zero/negative Sales or Quantity
iii)Confirm Cost of Sales is directionally sensible relative to Sales
iv)Convert Date to a proper date type; derive day-of-week, month, and week-of-year columns for later trend analysis


Step 4 :Elasticity & insights
i) For each of the 3 promo periods: calculate % change in quantity vs. % change in price relative to an adjacent non-promo baseline
ii) PED = %Δ Quantity ÷ %Δ Price; classify each as elastic (>1), inelastic (<1), or unit-elastic
iii)Compare gross profit % during promo vs. non-promo to judge whether promos help or hurt profitability
iv)Pull additional insights: seasonality (day-of-week/month patterns), volume trend over time, any outlier days
Step 3 : Derive metrics
METRICS
Daily sales price per unit = Sales ÷ Quantity Sold
Average unit sales price = mean (or period-weighted mean) of daily price/unit
Daily % gross profit = (Sales − Cost of Sales) ÷ Sales
Flag candidate promotional periods: look for sustained dips in price/unit paired with spikes in Quantity Sold vs. the surrounding baseline
Daily % gross profit per unit = (price/unit − cost/unit) ÷ price/unit
Select the 3 clearest, most defensible promo windows


Step 5 : Present & review
i)Build supporting visuals: price/volume trend line, GP% trend, a PED comparison chart across the 3 periods
ii) Summarize KPIs and the "better or worse on promo" conclusion in plain language
iii) Self-review all numbers and chart labels for consistency with the underlying data
iv) Export the final deliverable and submit